

Hotel Revenue Management System

A real-time analytics platform for optimizing hotel room pricing, automating KPI tracking, and maximizing revenue through data-driven insights.

Project Team

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Purpose

To optimize hotel room pricing and maximize revenue using real-time analytics and data-driven insights.

Automated KPI Tracking

Continuously monitor ADR, RevPAR, and Occupancy rates without manual intervention.

Streamlined Front-Desk Workflows

Responsive reservation systems for fast, accurate guest check-ins and room assignments.

Dynamic Market Pricing

Eliminate manual adjustments by deploying structural rate multipliers tied to live occupancy levels.

Problem Statement

Traditional hotel management relies on static pricing models that fail to react to real-time occupancy fluctuations and market demand.

Manual KPI Tracking

Difficulty tracking volatile metrics like ADR and RevPAR without automation.

Multi-Channel Data Gaps

Inefficient synchronization leads to overbookings or missed revenue opportunities.

Slow Response Times

Delayed reactions to guest cancellations and immediate check-in trends reduce revenue.

Goals

1

Automated KPI Dashboard

Centralized real-time audit of ADR, RevPAR, and Occupancy impact across all channels.

2

Smart Forecasting

Process historical records to generate tactical business projections and trend models.

3

Algorithmic Pricing

Deploy structural rate multipliers tied to instant system occupancy for maximum profitability.

Scope

In Scope

- Real-time KPI Dashboard tracking ADR, RevPAR, and Occupancy
- Instant reservation front-desk form for quick check-ins
- Dynamic pricing engine with structural rate multipliers
- Multi-channel API distribution pipelines

Out of Scope

- Standalone offline desktop application (fully web/cloud-based only)
- Third-party payment gateway integrations (mock/simulation only)
- Hardware-level keycard generation systems

Users



Hotel Administrators / Managers

Monitor revenue analysis, analyze booking channels, and configure baseline admin pricing multipliers.



Front-Desk Agents

Handle fast guest check-ins, process instant cancellations, and assign vacant inventory directly.



Executive Stakeholders

Review automated business projections, trend forecasting models, and macro performance statistics.

Functional Requirements

1

KPI Monitoring Module

Evaluate and display ADR, RevPAR, and live Occupancy rates simultaneously on the main dashboard.

2

Dynamic Pricing Engine

Automatically update structural rate multipliers based on current inventory status and room fill ratios.

3

Front-Desk Operations

Instant Reservation Form with validation for vacant room selection and primary guest name entries.

4

Data Integration Layer

Compile persistent live APIs across multi-channel distribution paths to prevent booking overlaps.

Non-Functional Requirements



Performance & Latency

Live metric updates must calculate and reflect changes within sub-second intervals upon front-desk actions.



Scalability

Securely manage concurrent distribution pipelines without database locks during high-traffic booking windows.



Usability & UI Design

Clear modular architecture with intuitive navigation: Dashboard, Revenue Analysis, Booking Channels, Pricing/Admin, and Logout.

Assumptions



Data Availability

The system assumes a reliable stream of baseline historical booking records is available to compute precise trend forecasting models.



Network Connectivity

Front-desk workstations maintain stable internet connectivity to process live API distribution pipelines without synchronization lag.



Inventory Accuracy

All active room statuses are updated exclusively through the application platform to maintain a single source of truth for occupancy calculations.

Acceptance Criteria

Scenario 1: Live Calculation Accuracy

Given a front-desk operator registers a new booking, **when** the reservation is confirmed, **then** the Occupancy Rate and corresponding RevPAR values must immediately recalculate on the main view.

Scenario 2: Algorithmic Pricing Adjustments

Given hotel occupancy surpasses a configured threshold, **when** a new query checks room availability, **then** the system must display the dynamically multiplied price tier automatically.

Scenario 3: Fault-Tolerant Inputs

Given the Instant Reservation Form is submitted, **when** mandatory parameters like "Primary Guest Name" are missing, **then** the interface must halt submission and present explicit input errors.